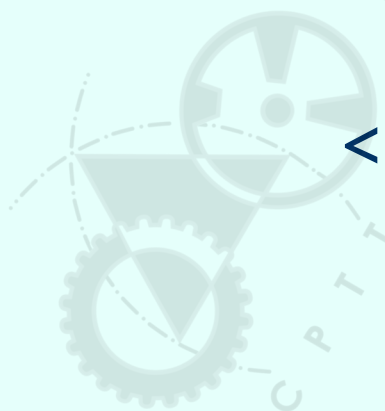


JP24July

Online Friendly Competition

Mobile
Applications
Development

<Module TW>



Module TW

Introduction

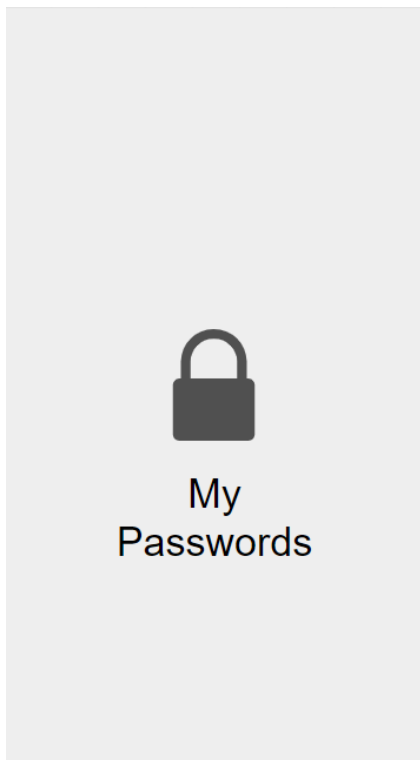
Please design a "Password Manager" program. Use built-in icons according to the page items and functional requirements to implement the "Password Manager" app's launch page, login page, registration page, password vault page, add password page, view password page, and edit password page.

Description of project and tasks

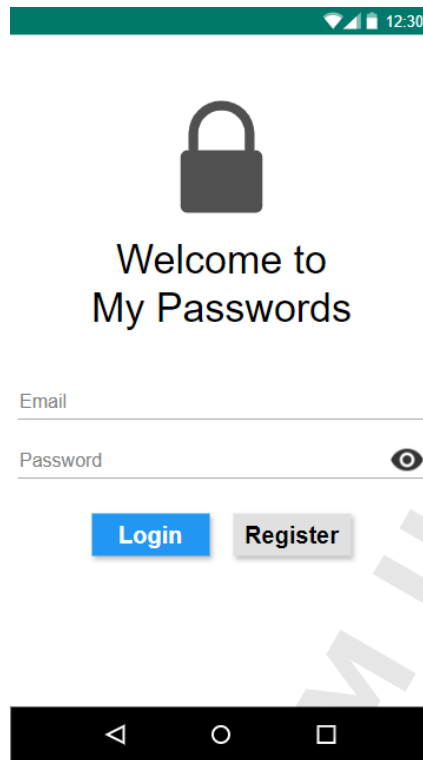
- The developer does not need to consider the aesthetics of the program interface and the position of the elements. The work can be inconsistent with the given wireframes, but the developer needs to properly consider the ease of use.
- The Test Project **MAY NOT** align with the provided wireframes.
- Competition time: 2.5 hours
- Marking emulator: Android – Pixel5 API 31, iOS – iPhone 8 Plus



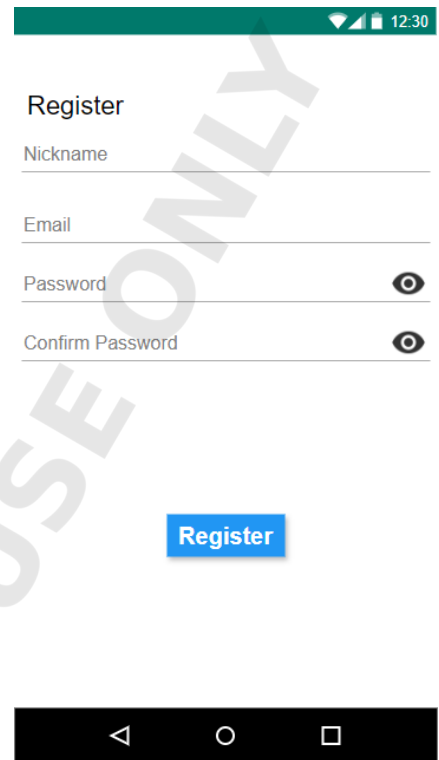
Application Wireframes



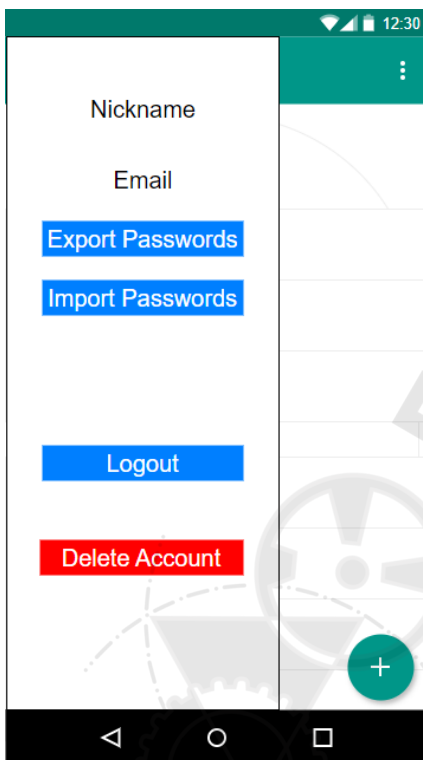
Launch Page



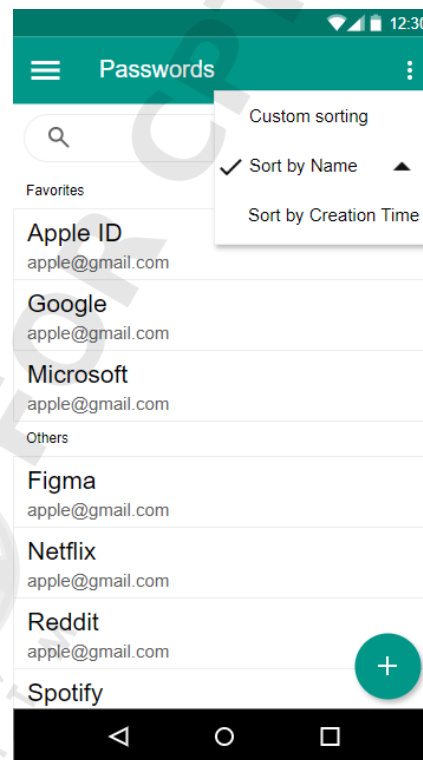
Login Page



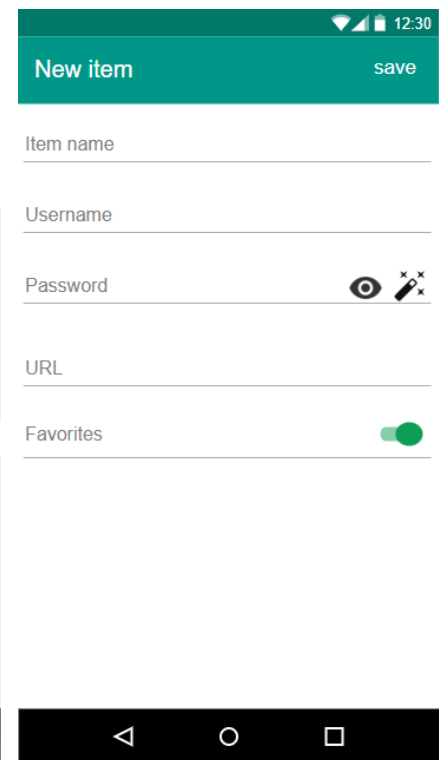
Registration Page



Drawer



Passwords Page



New Item Page

1. Launch Page

1. Elements include:
 - a. APP Icon  (lock).
 - b. APP Name.
2. Functional requirements:
 - a. Automatically navigate to the next page after 3 seconds.
 - i. Navigate to the login page if not logged in.
 - ii. Navigate to the **Passwords Page** if already logged in.

2. Login Page

1. Elements include:
 - a. Email input field.
 - b. Password input field.
 - c. Show password button.
 - d. Login button.
 - e. Register button.
2. Functional requirements:
 - a. The password input field must replace entered characters with *.
 - b. Pressing the icon  (visibility) on the right side of the password input field will temporarily show the entered characters, and releasing it will hide them.
 - c. When the login button is clicked, check the local SQLite database for an account and password that match the input field contents. If none match, display a login failure message.
 - d. Password vault data must be linked to account data. For example, Account A can only see passwords related to Account A, and logging in with Account B must not show Account A's information.
 - e. Clicking the register button navigates to the **Registration Page**.

3. Registration Page

1. Elements include:
 - a. Email input field.
 - b. Nickname input field.
 - c. Password input field.
 - d. Show password button.
 - e. Confirm password input field.
 - f. Register button.
2. Functional requirements:
 - a. If the Email or password contains whitespace characters, display a warning error message.

- b. If the Email exceeds 30 characters, display a warning error message.
- c. If the Email is not in email format, display a warning error message.
- d. If the password exceeds 15 characters, display a warning error message.
- e. Design the show password button.
- f. The password input field must replace entered characters with *.
- g. If the password input field and the confirm password input field do not match, display a warning error message.
- h. When the register button is clicked, check if there is an account with the entered Email in the current database.
 - i. If an account exists, display the warning error message "Email account already exists!".
 - ii. If no account exists, create the account and navigate to **Passwords Page**.

4. Drawer

1. Elements include:
 - a. Email field.
 - b. Nickname field
 - c. Export Passwords button.
 - d. Import Passwords button.
 - e. Logout button.
 - f. Delete Account button.
2. Functional requirements:
 - a. Display the currently logged-in Email.
 - b. Display the Nickname of the currently logged-in Email.
 - c. When the export passwords button is clicked, show a filename input field. After entering the filename, convert all password vault information of the current account into a JSON file and save it to the phone's Temporary Directory.
 - d. When the import passwords button is clicked, filter JSON files from the Temporary Directory and display a list. Click on a JSON file to import the Passwords.
 - e. When the logout button is clicked, display a confirmation dialog to confirm the logout.
 - i. Click confirm to clear Shared Preferences and navigate to the login page.
 - ii. Click cancel to return to the Drawer.
 - f. When the delete account button is clicked, display a dialog with the following:
 - i. Password input field.
 - ii. Confirm password input field.
 - iii. Confirm button.
 - iv. Cancel button.
 - v. Click confirm to check if the password is correct. If correct, proceed to delete the account and related password items.

5. Passwords Page

1. Elements include:
 - a. Page Title: Passwords.
 - b. Sort button  (more_vert), located on the right side of the title bar.
 - c. Search input field.
 - d. Passwords list, containing sections for "Favorites" and "Others", with multiple password entries in each section.
 - e. New item button, icon represented by  (add).
2. Functional requirements:
 - a. Passwords list data source is SQLite, requiring database design.
 - b. Passwords list must distinguish between "Favorites" and "Others", with "Favorites" section pinned at the top.
 - c. Clicking on a password entry switches to the Item Details.
 - d. Implement search functionality using a SearchBar, searching by password item name and username.
 - e. Clicking the new item button switches to the New Item Page.
 - f. Sort button:
 - i. The sorting of Passwords list will be based on the selected sorting method, applying to both "Favorites" and "Others" sections.
 - ii. Clicking the sort button displays a menu with options: "Custom Sorting", "Sort by Name", "Sort by Creation Time".
 - iii. The selected option should display a  (check) icon.
 - iv. Clicking "Sort by Name" or "Sort by Creation Time" should display either an ascending  (arrow_drop_up) or descending  (arrow_drop_down) icon next to the selected option, defaulting to ascending order.
 - v. If the selected sorting order is displayed as ascending, clicking again will change it to descending, and vice versa.
 - vi. For custom sorting, initially sort by creation date in ascending order. Users can adjust the order by long-pressing and dragging items. When selecting custom sorting again, sort according to the previous custom order.
 - vii. Store sorting method and order (ascending/descending) in Shared Preferences; store custom sorting order in the database.

6. New Item Page

1. Elements include:
 - a. Page Title: New Item.
 - b. Item Name input field.
 - c. Username input field.

- d. Password input field.
- e. URL input field.
- f. Favorites Switch.
- g. save button.
- h. Password Generator.
 - i. Length Slider, minimum value 5, maximum value 64, default 10.
 - ii. Uppercase Switch, default True.
 - iii. Lowercase Switch, default True.
 - iv. Numbers Switch, default True.
 - v. Symbols (!@#\$%^&*) Switch, default True.
 - vi. Custom Characters input field, length 20.
 - vii. Custom Characters Switch, default False.
 - viii. Password display area.
 - ix. Regenerate Password button.
 - x. Copy Password button.
 - xi. confirm button.

2. Functional requirements:

- a. All input fields are mandatory with a maximum length of 100 characters.
- b. The password input field must have two buttons:
 - i. Show/Hide Password button, toggled with  (visibility) and  (visibility_off) icons.
 - ii. Generate Password button, using  (auto_fix_high) icon, which opens the password generator.
- c. The URL input field needs to validate if the format is a valid URL.
- d. Clicking the save button saves data to the database. The value of the password input field must be stored in the database.
- e. Clicking the Cancel button switches to the Passwords Page.
- f. Password Generator:
 - i. Upon opening the password generator for the first time, generate a password based on default conditions and display it in the password display area. For subsequent uses, load user-defined conditions from Shared Preferences and generate passwords accordingly.
 - ii. Behind the password display area, there should be two buttons:
 - Copy Password button, using  (content_copy) icon.
 - Regenerate Password button, using  (refresh) icon.
 - iii. When conditions change, immediately generate a new password.
 - iv. Generated passwords must adhere to user-defined conditions. If Numbers or Symbols switches are True, include at least one number or one symbol.

- v. Display alphanumeric characters in black font, numbers in blue font, and symbols in green font.
- vi. Clicking the Regenerate Password button generates a new password based on the current conditions.
- vii. Clicking the Confirm button closes the password generator, fills the generated password into the password input field, and saves user-defined conditions to Shared Preferences.
- viii. Clicking the Copy Password button copies the generated password to the clipboard.

7. Launch Page

1. Elements include:
 - a. Page Title: Item Details.
 - b. Item Name.
 - c. Add to Favorites button.
 - d. Username.
 - e. Copy Username button.
 - f. Password.
 - g. Show/Hide Password button.
 - h. Copy Password button.
 - i. URL.
 - j. Creation/Update Time.
 - k. edit button, located on the right side of the page title.
 - l. Delete button, located at the bottom of the page.
2. Functional requirements:
 - a. Behind the Item Name display area, there should be an Add to Favorites button, represented by  (star) or  (star_border) to indicate whether the item is favorited or not.
 - b. Clicking the Add to Favorites button should toggle the favorite status of the item.
 - c. Behind the Username, there should be a Copy Username button with the  (content_copy) icon.
 - d. Clicking the Copy Username button should copy the username to the clipboard.
 - e. Behind the Password, there should be two buttons:
 - i. Show/Hide Password button, toggled with  (visibility) and  (visibility_off) icons to toggle password visibility.
 - ii. Copy Password button, using the  (content_copy) icon. Clicking it should copy the password in plaintext to the clipboard.
 - f. When the password display is hidden, regardless of password length, display eight asterisks (*).
 - g. When showing the password, display alphanumeric characters in black font, numbers in blue font, and symbols in green font.

- h. Clicking the URL should open the corresponding website in the default browser.
- i. Clicking the edit button should navigate to the Item Editing Page.
- j. Clicking the Delete button should prompt a confirmation dialog "Confirm delete this password item". Upon confirmation, delete the item from the database. If canceled, close the confirmation dialog.

8. Item Editing Page

1. Elements include:
 - a. Page Title: Item Editing.
 - b. Other elements same as the New Item Page.
2. Functional requirements:
 - a. Load the corresponding data from the database and populate the respective input fields.
 - b. Other functional requirements are the same as those for the New Item Page.

